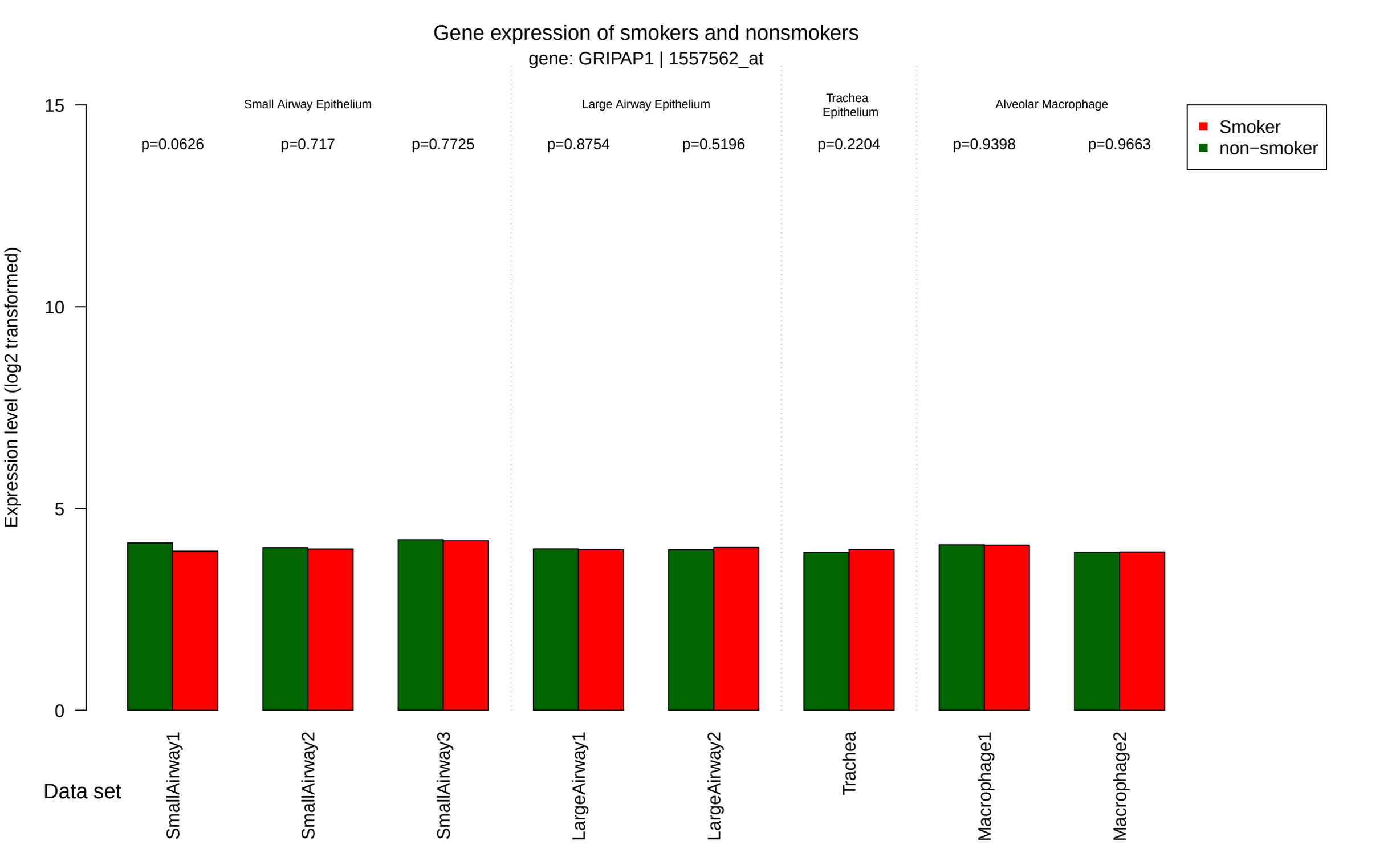
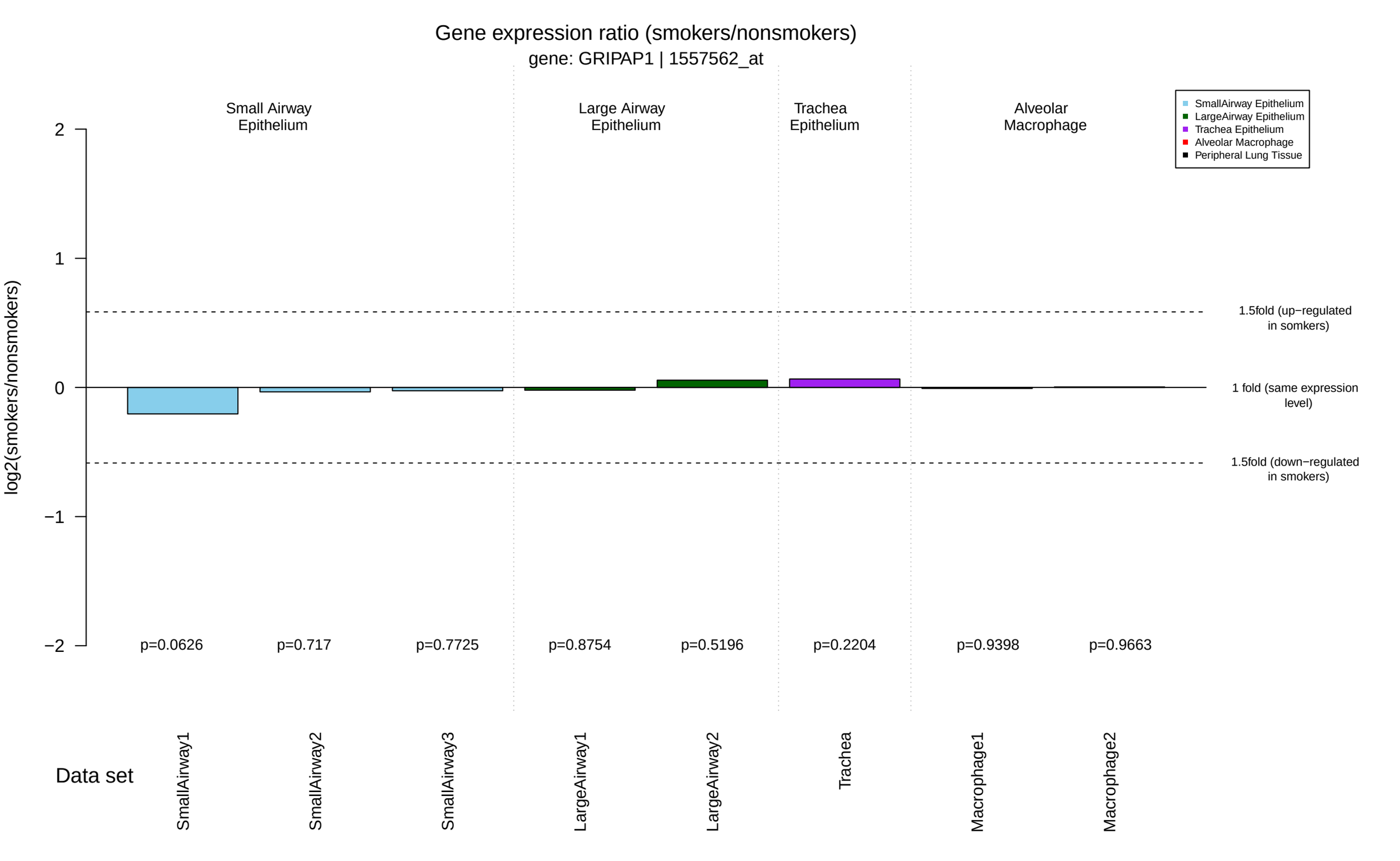
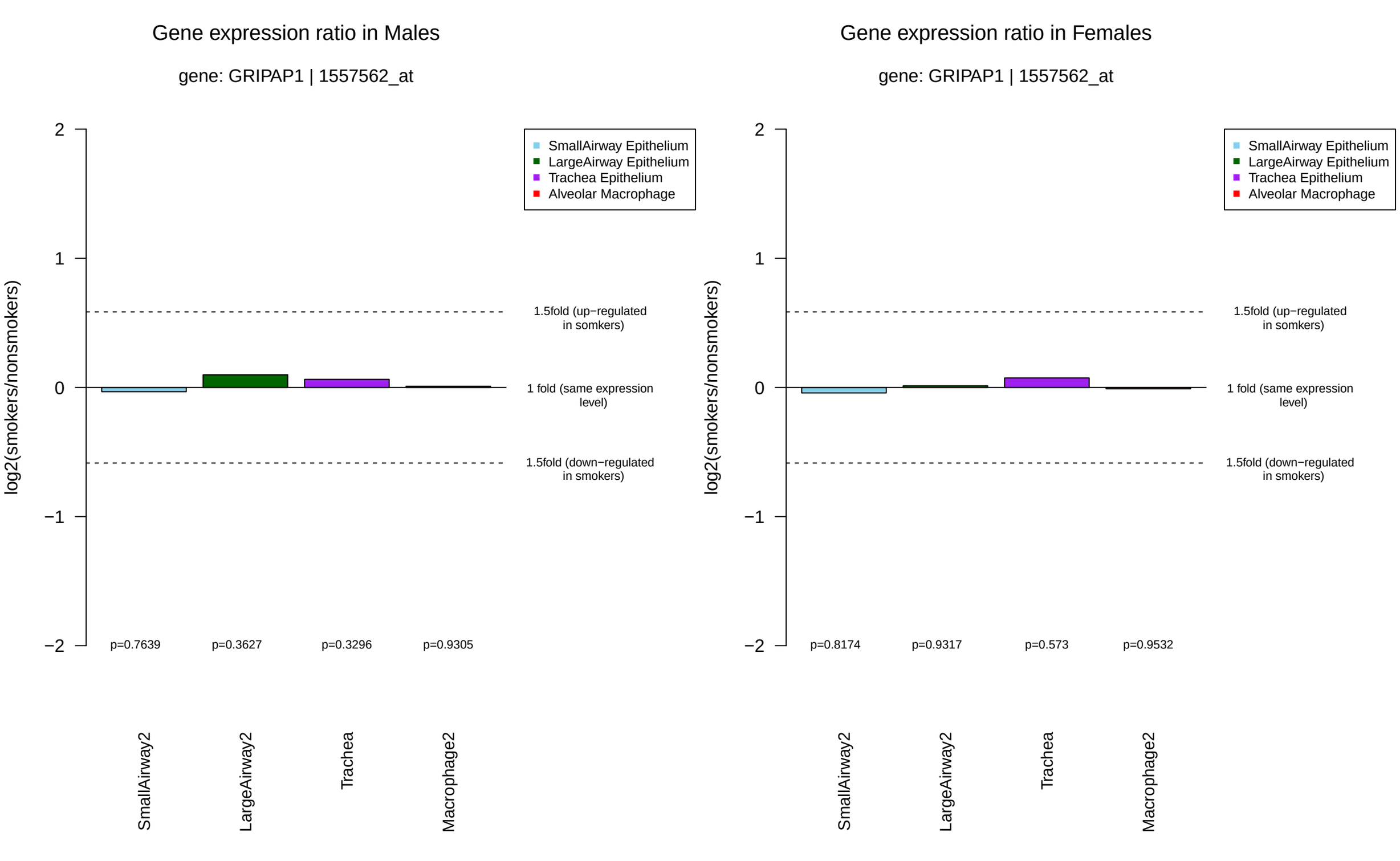
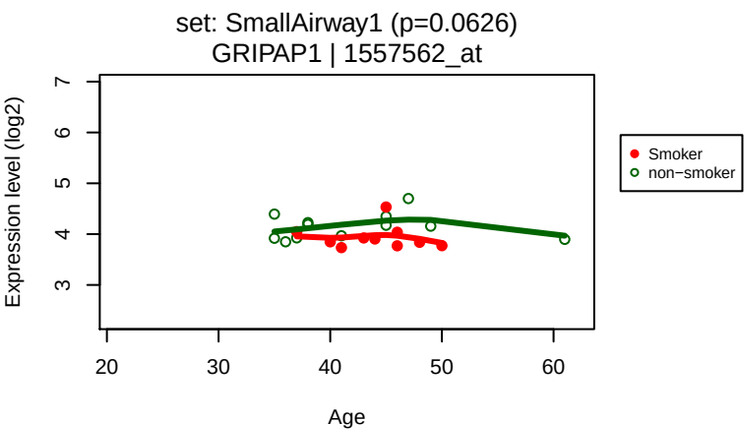
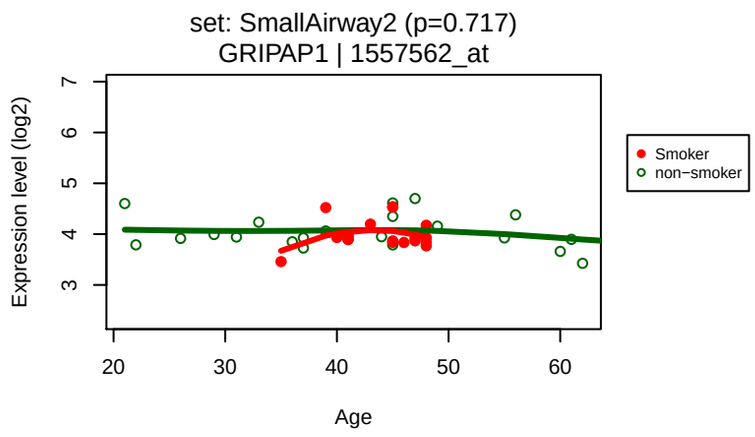
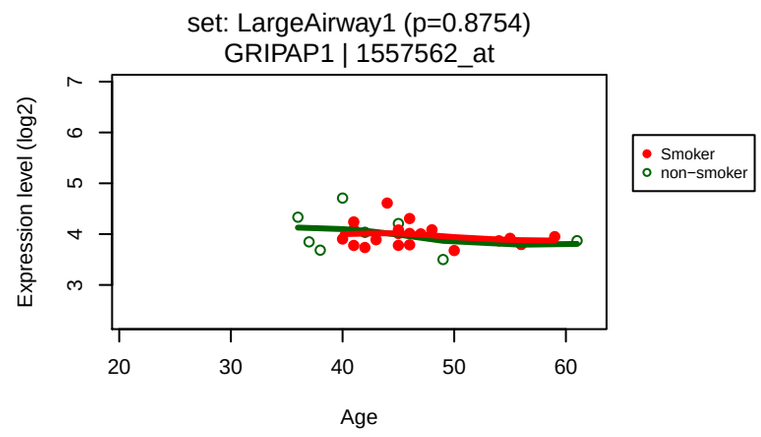

Gene expression of smokers and nonsmokers
gene: GRIPAP1 | 1557562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0626
p=0.717
p=0.7725
p=0.8754
p=0.5196
p=0.2204
p=0.9398
p=0.9663
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

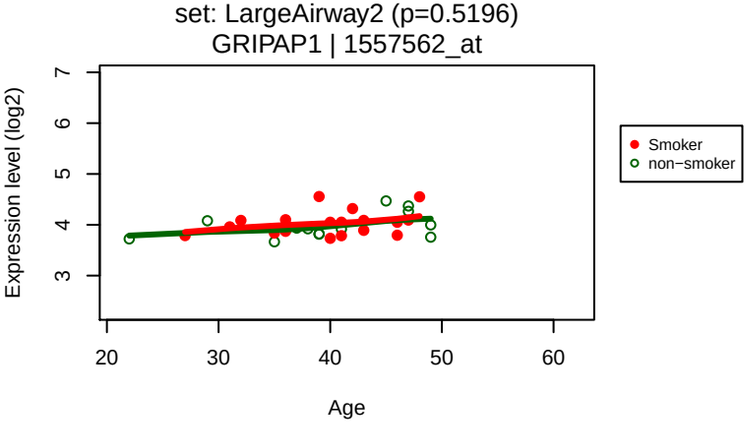

Gene expression ratio (smokers/nonsmokers)
gene: GRIPAP1 | 1557562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0626
p=0.717
p=0.7725
p=0.8754
p=0.5196
p=0.2204
p=0.9398
p=0.9663
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

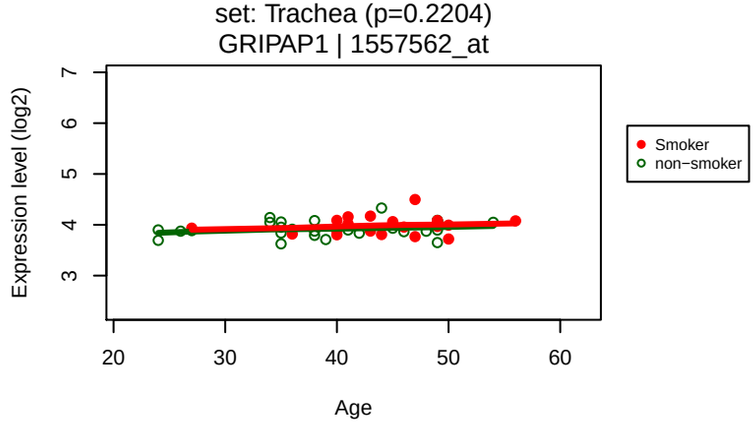

Gene expression ratio in Males
gene: GRIPAP1 | 1557562_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7639
p=0.3627
p=0.3296
p=0.9305
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIPAP1 | 1557562_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8174
p=0.9317
p=0.573
p=0.9532
SmallAirway2
LargeAirway2
Trachea
Macrophage2

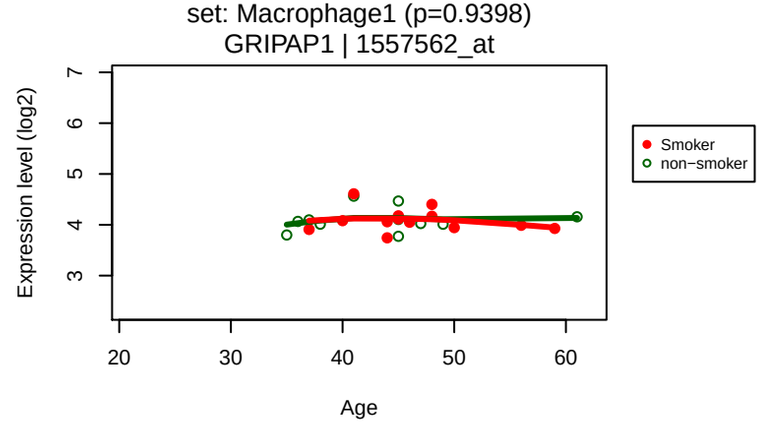

set: SmallAirway1 (p=0.0626)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker

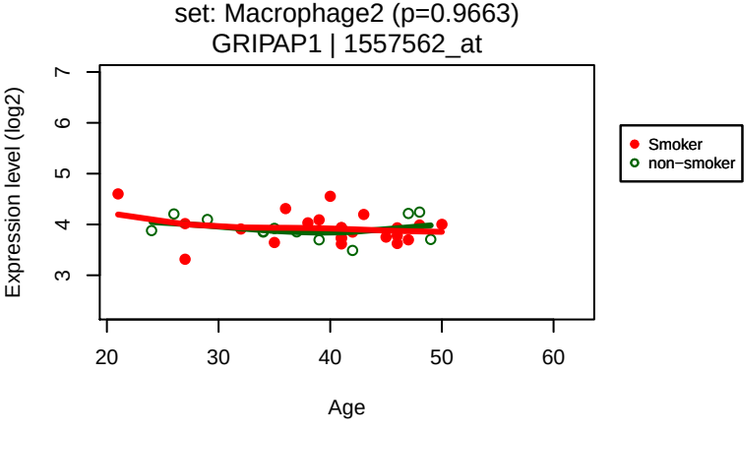

set: SmallAirway2 (p=0.717)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker

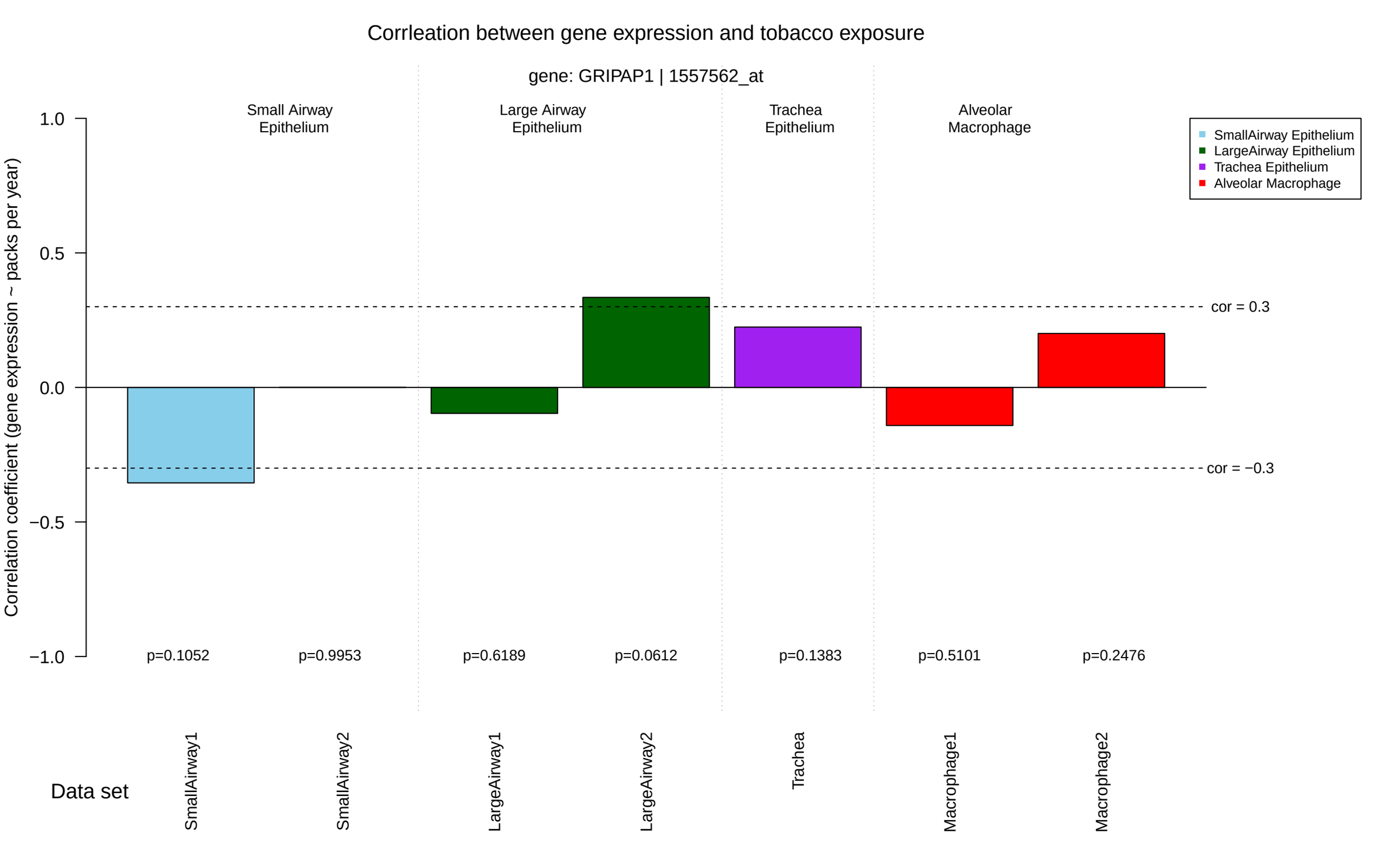

set: LargeAirway1 (p=0.8754)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5196)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2204)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9398)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9663)
GRIPAP1 | 1557562_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIPAP1 | 1557562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1052
p=0.9953
p=0.6189
p=0.0612
p=0.1383
p=0.5101
p=0.2476
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2